



**THE STATE UNIVERSITY OF ZANZIBAR
SCHOOL OF COMPUTING AND COMMUNICATION TECHNOLOGIES
DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION
TECHNOLOGY**

PROPOSAL

e-Land Verification and Management System (e-LVMS)

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INTRODUCTION

The e-Land Verification and Management System is a secure digital system designed to make land record management and verification easier. The system allows authorized land officers from the Ministry of Lands to register, manage, and verify land information. This helps to make sure that land records are correct, official, and reliable.

The system also allows land surveyors to submit survey information, including GPS points. These points are used to show the land location on a digital map.

After the land verification process is completed, the system generates a unique Area Verification Number (AVN) for the land owner. The owner can use the AVN to search for and view verified land information through the public dashboard.

To protect the owner's information, the system generates a temporary secret code when an AVN search is made. The secret code is sent to the owner's registered email address and expires after two minutes. This helps prevent unauthorized people from accessing the owner's land information.

The system also supports land registration, surveyor assignment, land verification, payment management, staff management, and report generation. Overall, the system provides a more organized and secure way to manage and access verified land records.

CURRENT SITUATION

At the moment, land information is mainly handled manually. Land owners have to visit land offices to register or verify their land. Most land records are kept on paper, which makes the process slow, difficult to manage, and more likely to have errors.

PROBLEM STATEMENT

- Traditional land records are mostly paper-based, slow, and prone to errors.
- Land registration and verification take a lot of time and are not efficient.

- There is a risk of fraud, duplicate ownership, and land disputes.
- Land owners cannot easily access official and verified land information.
- There is no single digital system for tracking and verifying land records.
- A secure and transparent digital system is needed to store and manage verified land records.
- Land owners need a safe way to view their verified land details using a unique Area Verification Number (AVN).

MAIN & SPECIFIC OBJECTIVES

Main Objective

- ☐ To develop a secure and reliable digital system for managing, verifying, and accessing land ownership records using GIS for accurate land measurements.

Specific Objectives

- ☐ To digitize land records and minimize reliance on paper-based systems.
- ☐ To allow authorized land officers to register and verify land ownership accurately.
- ☐ To utilize GIS technology for precise land measurement from all angles.
- ☐ To generate a unique Area Verification Number (AVN) for each verified land.
- ☐ To provide land owners with easy and safe access to their land information through a public dashboard.
- ☐ To prevent fraud, duplication, and errors in land registration.
- ☐ To improve transparency, efficiency, and reliability in land management processes.

TECHONOLGY AND TOOLS

- Spring Boot (Java) – Used to handle the backend logic, land verification, security, and AVN generation.
- React.js– Used to build the user interface for land officers, land surveyors, and the public dashboard.
- MySQL – Used as the main database to securely store land records and other system information.
- REST APIs– Used to allow communication between the frontend and backend.
- GIS Technology– Used to display land locations and GPS points on a digital map.

TOOLS USED

- **Visual Studio Code** – Used to write and edit the system code.
- **Postman** – Used to test the REST APIs.
- **Swagger** – Used to test and check the backend APIs.
- **GitHub** – Used to keep a backup of the project code.
- **phpMyAdmin** – Used to manage and view the MySQL database.
- **Web Browser** – Used to run and test the system.

EXPECTED OUTPUT

- **Public Dashboard** – Allows land owners to search and view verified land information using the Area Verification Number (AVN).
- **Land Officer Panel** – Allows authorized land officers to register, manage, and verify land records.

- **Login System** – Provides secure access for admins, land officers, and land surveyors.
- **Database** – Stores land records, owner information, survey information, AVN, and other system data.
- **GIS Maps** – Displays land locations and GPS points on a digital map.
- **Area Verification Number (AVN)** – A unique number generated for verified land.
- **Reports** – Allows land officers to generate reports about land records and verification.
- **Security Features** – Protects system and user information through authentication, password encryption, and a temporary secret code.